

BACKGROUND	
Title/Citation	Rickels K, Pollack MH, Feltner DE, et al. Pregabalin for Treatment of Generalized Anxiety Disorder: A 4-Week, Multicenter, Double-blind, Placebo-Controlled Trial of Pregabalin and Alprazolam. Arch Gen Psychiatry. 2005;62(9):1022–1030. doi:10.1001/archpsyc.62.9.1022 https://jamanetwork.com/journals/jamapsychiatry/fullarticle/1108404
Background	This article detailed report on a 4-week, multicenter, double-blind, placebo-controlled clinical trial that evaluates the efficacy, safety, and tolerability of pregabalin in the treatment of generalized anxiety disorder. The study also compares pregabalin with alprazolam, a commonly prescribed anxiolytic agent, to assess pregabalin's anxiolytic efficacy.
GENERAL STUDY OVERVIEW	
Purpose	To assess the anxiolytic efficacy of pregabalin in patients with generalized anxiety disorder.
Funding	This study was supported by Pfizer Inc, New York, NY
Design	Double-blind, placebo-controlled, active-comparator trial. Patients were randomized to 4 weeks of treatment with 3 different dose of pregabalin and 1.5 mg Alprazolam dose or placebo.
METHODS	
Inclusion criteria	Male or female outpatients who were 18 years or older, met the <i>DSM-IV</i> criteria for GAD and had screening and baseline scores of 20 or greater on the Hamilton Anxiety Rating Scale (HAM-A) and 9 or greater on the Covi Anxiety Scale were eligible for enrollment.
Exclusion criteria	Raskin Depression Scale 28 score of greater than 7; Fertile woman or having a positive pregnancy test result, not using a medically accepted contraceptive, or currently nursing; Current or past history of bipolar, schizophrenic, psychotic, or factitious disorder and dementia; substance use disorder; Heart problems; Current or past history of a seizure disorder or under treatment.
Intervention	After a 1-week drug-free screening period, 454 patients were randomized to receive fixed doses of Pregabalin (300 mg/d, 450 mg/d, or 600 mg/d), Alprazolam (1.5 mg/d), or Placebo for 4 weeks. Pregabalin dosing was titrated over the first week to the assigned dose, administered thrice daily. The study included a 1-week taper period and a 1-week follow-up to monitor discontinuation effects.
Primary outcome	The primary measure was the change in total HAM-A score from baseline to endpoint. HAM-A assessments occurred at screening, baseline, weeks 1–4, and during the taper period for some patients.
Secondary outcomes	Secondary efficacy measures included the 17-item clinician-rated HAM-D at screening, baseline, and week 4, and the investigator-rated CGI-I at weeks 1–4. The psychic and somatic anxiety factors of HAM-A were assessed, along with anxiety/worry and tension, as these are central to GAD diagnosis
Procedures	Patients were randomized to 4 weeks of treatment with pregabalin, 300 mg/d (n = 91), 450 mg/d (n = 90), or 600 mg/d (n = 89); alprazolam, 1.5 mg/d (n = 93); or placebo (n = 91) in double blinded treatment with follow up and tapering down the medications to avoid withdrawal side effect.
Statistical analyses	All statistical analyses were performed using SAS statistical software for the intent-to-treat (ITT) population, composed of all randomized patients who received at least 1 dose of study medication. A sample size of 97 evaluable patients per treatment group would provide 85% power to detect a mean difference of 3.5 (SD, 7) in the HAM-A score between placebo and pregabalin (300, 450, or 600 mg/d), with an experiment wise α level of .05. Last-observation-carried-forward (LOCF) analysis was used on all primary and secondary outcome measures for the planned analyses.
RESULTS	
Baseline characteristics	A total of 696 patients were screened, total of 454 patients were randomized, with similar baseline demographic and clinical characteristics across treatment groups.

Summary of outcomes	For the primary end point, change in HAM-A score, the LOCF end point analysis showed that all 3 doses of pregabalin and alprazolam had significantly greater efficacy than placebo with measuring criteria of a CGI-I score of 2 or less (much or very much improved) ($\chi^2_4 = 17.1$, $P < .01$) and a 50% or greater reduction from baseline in the HAM-A total score ($\chi^2_4 = 14.0$, $P < .01$). Responder rates at week 4 – Scale total score from baseline to end point was highest for pregabalin, 600 mg (12.9 ± 1.9); followed by reductions of 10.6 ± 1.6 for pregabalin, 450 mg; 9.1 ± 1.9 for pregabalin, 300 mg; 7.2 ± 1.7 for alprazolam; and 7.2 ± 1.7 for placebo. Only pregabalin, 600 mg, demonstrated significantly greater improvement in work productivity than placebo ($P < .03$), while pregabalin, 450 mg, did so at a statistical trend level ($P < .14$).
AUTHORS' DISCUSSION & CONCLUSIONS	
Author's Discussion	Pregabalin and alprazolam produced a significantly greater reduction in mean $\pm$ SE HAM-A total score at last-observation-carried-forward end point compared with placebo (-8.4 ± 0.8): pregabalin, 300 mg (-12.2 ± 0.8, $P < .001$), 450 mg (-11.0 ± 0.8, $P = .02$), and 600 mg (-11.8 ± 0.8, $P = .002$), and alprazolam (-10.9 ± 0.8, $P = .02$). By week 1 and at last-observation-carried-forward end point, the 3 pregabalin groups and the alprazolam group had significantly ($P < .01$) improved HAM-A psychic anxiety symptoms compared with the placebo group. Compared with the placebo group, HAM-A somatic anxiety symptoms were also significantly ($P < .02$) improved by the 300- and 600-mg pregabalin groups, but not by the 450-mg pregabalin (week 1, $P = .06$; week 4, $P = .32$) and the alprazolam groups (week 1, $P = .21$; week 4, $P = .15$).
Author's conclusions	Pregabalin was significantly more efficacious than placebo for the treatment of psychic and somatic symptoms of generalized anxiety disorder and was well tolerated by most study patients.
PRESENTER'S DISCUSSION & CONCLUSIONS	
Study strengths	Appropriately powered, designed to minimize confounding (with blinding, randomization, ITT analysis, and inclusion/exclusion criteria).
Study limitations	Lyrica was manufactured by Pfizer, and it is more costly than Alprazolam or Xanax which was originally manufactured by UpJohn and later acquired by Pfizer.
Conclusion and Application to patient care	This study showed a significant result to show the use of Pregabalin to use in the GAD treatment than placebo or current Guidelines treatment with Benzodiazepines.

Background on Generalized Anxiety Disorder (GAD)

Generalized anxiety disorder is a prevalent, chronic psychiatric condition characterized by excessive anxiety and worry, often accompanied by somatic symptoms such as tension and autonomic manifestations. It typically has an onset in late adolescence or early adulthood and a duration often exceeding ten years. GAD is prevalent in primary care settings, with a 12-month prevalence estimated between 3% and 5% and is associated with significant functional impairment and healthcare utilization.

Pharmacology of Pregabalin

Pregabalin is a novel compound structurally related to GABA but does not act on GABA receptors or inhibit serotonin or norepinephrine reuptake. Instead, it binds with high affinity to the $\alpha 2$ - δ subunit of voltage-dependent calcium channels in the brain and spinal cord, modulating the release of excitatory neurotransmitters. Its pharmacokinetic profile includes rapid absorption (peak concentration at approximately 1 hour), linear kinetics, minimal protein binding, and primary renal excretion. These properties suggest a favorable drug-drug interaction profile.